

❓ PHASE 0 — Foundations

Computer Basics

- How web works (client–server)
- HTTP vs HTTPS
- Request / response cycle
- REST concept

Tools

- Git & GitHub
- Maven / Gradle
- IntelliJ IDEA / Eclipse

❑ Do NOT skip this phase

❑ PHASE 1 — Core Java (BACKBONE)

You must master **logic** + **OOP**, not syntax.

Core Java Topics

- JVM, JRE, JDK
- Data types, control flow
- OOP:
 - Encapsulation
 - Inheritance
 - Polymorphism
 - Abstraction
- Constructors
- `static`, `final`, `this`, `super`
- Packages & access modifiers
- Exception handling
- Collections Framework
- Generics
- Multithreading
- File I/O
- Lambda expressions
- Streams API

❑ If Core Java is weak → **everything collapses**

☐ PHASE 2 — SQL & RDBMS

Learn SQL deeply

- DDL, DML, DCL
- Joins (INNER, LEFT, RIGHT)
- Subqueries
- Indexes
- Normalization
- Transactions
- ACID properties

Database

- MySQL or
- PostgreSQL

☐ Java without SQL = **half engineer**

☐ PHASE 3 — JDBC (Java ↔ Database)

- JDBC architecture
- Drivers
- Connections
- PreparedStatement vs Statement
- ResultSet
- Connection pooling
- CRUD operations

☐ This teaches **how backend actually talks to DB**

☐ PHASE 4 — Web Basics (Frontend for Java Devs)

You don't need fancy UI — only **functional UI**.

Frontend Basics

- HTML5

- CSS3
- Basic JavaScript
- Forms
- DOM basics

☐ Enough to connect backend to UI

☐ **PHASE 5 — Java Web (Servlets & JSP)**

Java Web Stack

- Servlet lifecycle
- RequestDispatcher
- Sessions & Cookies
- JSP directives
- Expression Language
- JSTL
- MVC pattern

☐ This is **core Java web understanding**

☐ **PHASE 6 — Spring Framework (CRITICAL)**

This is where you become **industry-ready**.

Spring Core

- IoC
- Dependency Injection
- Bean lifecycle
- Annotations

Spring MVC

- Controllers
 - Request mappings
 - Form handling
 - Validation
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☐ PHASE 7 — Spring Boot (MANDATORY)

Spring Boot Topics

- Auto-configuration
- REST APIs
- DTOs
- Exception handling
- Logging
- Pagination
- Filtering
- Validation

☐ Spring Boot is **non-optional** in 2025

☐ PHASE 8 — JPA & Hibernate (DATA LAYER)

- ORM concepts
- Entities
- Relationships
- JPQL
- Criteria API
- Lazy vs Eager loading
- Transactions
- Caching

☐ Without ORM knowledge → **no enterprise job**

☐ PHASE 9 — Security

- Spring Security
 - Authentication vs Authorization
 - JWT
 - Role-based access
 - Password hashing
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☐ PHASE 10 — Advanced Backend

- Microservices basics
 - REST best practices
 - API versioning
 - Caching with **Redis**
 - Messaging (Kafka / RabbitMQ – optional)
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☐ **PHASE 11 — DevOps & Deployment**

- Docker basics
 - CI/CD concepts
 - AWS basics
 - Linux basics
 - Nginx
 - Deploy Spring Boot app
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☐ **PHASE 12 — Projects (MOST IMPORTANT)**

You must build:

1. CRUD App
2. REST API project
3. Auth-based system
4. Full-stack project (Java + UI + DB)

☐ No projects → **no credibility**